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# Cerebrovascular and metabolic changes during the delayed vasospasm following experimental subarachnoid hemorrhage in baboons, and treatment with a calcium antagonist

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A model has been designed in baboons for simulating the clinical situation during the late phase of vasospasm in patients with subarachnoid hemorrhage (SAH). A total amount of 14–33 ml autologous blood was injected into the cisternal system on 3 occasions in the course of 4 days. Neurological symptoms were seen, and the mortality rate was 29%. Angiography 3 days after the last injection showed arterial vasoconstriction amounting to 23% in the vertebro-basilar system, and 11% (right) and 18% (left) in the carotid system. Cerebral blood flow (CBF) measured by the intra-arterial $^{133}Xe$ technique and the cerebral metabolic rate of oxygen ($CMRO_2$) were reduced by 18% and 11%, respectively. The hypercapnic CBF response was significantly impaired, from a mean of 3.90 ml/100 g/min to 1.72 ml/100 g/min of flow increase for each mm Hg elevation of $p_aCO_2$. Autoregulation, tested by administration of angiotensin II, was also significantly affected as evidenced by a pressure-dependent increment of CBF during hypertension in 5 out of 7 animals tested. The impaired autoregulation was reflected in the autoregulatory index, which in the whole group increased from 0.06 ml/100 g/min for each mm Hg increase in MABP in the pre-SAH animals to 0.29 ml/100 g/min per mm Hg post-SAH. Treatment with the calcium antagonist, nimodipine (0.5 μg/kg/min i.v. during 45 min), enhanced CBF significantly by 17% before experimental SAH, whereas after SAH the effect was slight and did not reach statistical significance; $CMRO_2$ was not significantly affected in either group. Intravenous nimodipine combined with hypertension resulted in a marked increase in the autoregulatory index to 1.58 ml/100 g/min per mm Hg in pre-SAH animals and a less pronounced increment to 0.58 ml/100 g/min per mm Hg following experimental SAH. The beneficial effect of nimodipine reported in SAH patients is therefore, in view of our findings, more likely due primarily to a protective mechanism at the cellular level than to an influence on the vascular bed.

## INTRODUCTION

One of the most important aspects of SAH[1] is the development of progressive cerebral ischemia secondary to delayed arterial spasm in the brain. The associated clinical deterioration usually starts rather insidiously at the end of the first week after the bleeding, only to progress to more serious ischemic manifestations during the second week. Cerebral angiography during the second week after the bleed reveals signs of vasospasm in about 60% of all patients with SAH[86,130]. An increased circulation time has been noted angiographically in association with the spasm[48] and CBF studies have demonstrated an impairment of the cerebral circulation[39,59,64]. Computorized tomography has provided striking evidence for a correlation between neurological deficits, vasospasm, and ischemia[40,41,106]. There is a significant de-

*Abbreviations:* SAH, subarachnoid hemorrhage; CBF, cerebral blood flow; $CMRO_2$, cerebral metabolic rate of oxygen; SSP, sagittal sinus pressure; EEG, electroencephalogram; CSF, cerebrospinal fluid; MABP, mean arterial blood pressure.

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crease in $CMRO_2$ in patients with SAH as compared to normal controls[52,77].

However, a discrepancy is sometimes seen between the clinical condition, the angiographically determined vasospasm, and the CBF and $CMRO_2$. Thus, it is possible for patients to have clear-cut vasospasm on the angiogram without neurological dificits or any significant reduction in CBF. The ischemic potential of vasospasm may therefore also be related to other factors. Possible cerebrovascular factors include altered reactivity to carbon dioxide[93,127], impaired autoregulatory response to changes in blood pressure[34,94,96], and the relationship between CBF and brain metabolism[98].

There is at present no well-established effective treatment for the delayed ischemia secondary to vasospasm. Attempted therapeutic principles of current interest include hypervolemia along with hypotension[66,68,69], and treatment with calcium entry blockers, usually referred to as calcium antagonists[1,7,76,112].

We have designed an SAH model in baboons[103], in which repeated injections of autologous blood induces angiographic signs of vasospasm, with the aim of simulating the clinical situation during the late phase of vasospasm in patients with SAH. In the present study, this model has been evaluated with regard to the degree of vasoconstriction and associated changes in CBF, $CMRO_2$, and SSP, together with manifest neurological and gross neuropathological changes. The response to hypercapnia and to increased systemic blood pressure were studied following the experimental SAH. Attempts were made to ameliorate the cerebrovascular effects of SAH by intravenous infusion of the calcium uptake blocker, nimodipine, during which also the autoregulatory response was evaluated.

## MATERIALS AND METHODS

### *Anesthesia*

The material comprised 25 adult baboons (*Papio cynocephalus*) of either sex, weighing between 8.9 and 21.8 kg. Anesthesia was initiated with phencyclidine hydrochloride (Sernylan, Bio-Ceutic; 1 mg/kg i.m.) followed by thiopental sodium (Pentothal Natrium, Abbott; 7.5 mg/kg i.v.). The animals were intubated endotracheally and the anesthesia was maintained with 70% nitrous oxide and 30% oxygen. Respiration was controlled in a semi-open circuit using a servo-ventilator (Siemens-Elema 900). Muscular relaxation was achieved initially with suxamethonium chloride (Celocurin-klorid, Vitrum; 25 mg i.v.) and then maintained with 50 mg suxamethonium chloride intramuscularly every half hour. During the surgical procedures, 0.5% fluothane (Halothane, ICI) was added to the gas mixture. The fluothane was switched off after completion of surgery, and the animals were allowed to stabilize for about 90 min before the CBF measurements and the angiography were performed. For these procedures, a continuous intravenous infusion of phencyclidine (0.5 mg/kg/h) was given. In some instances, though, the rate of infusion was modified according to the activity of the EEG (see below).

### *Animal preparation*

The various steps in the complete experimental protocol are illustrated in Fig. 1. Of the original 25 animals, one was not injected with blood but instead received 2 ml cardiac green into cisterna magna as a marker to evaluate the distribution in the subarachnoid space. Seven animals died before the final angiography and the blood flow measurement could be performed.

*Stage I.* The animals were anesthetized as above, and 3-vessel angiography (bilateral common carotid and unilateral vertebral artery) was performed. Following this, the head of the animal was fixed in a stereotaxic frame and 8 extradural silver electrodes were inserted into the bone of the vertex for long-term use[21]. In the same session, the temporal muscle on one side was extirpated in order to eliminate errors in CBF measurements due to isotope clearance from the extracranial tissue[42]. Additionally, a small drill hole was made in the midline, just above the confluence sinuum, for later insertion into the superior sagittal sinus of a catheter, which was fixed in the drill hole with dental cement.

*Stage II.* Three to 7 days later the animals were again anesthetized. The bifurcation of the right common carotid artery was exposed. A catheter with an outer diameter of 0.9 mm was inserted into the linguo-facial trunk with the aid of an operation microscope for $^{133}Xe$ injections. The main trunk of the external carotid artery was clamped at its base using a

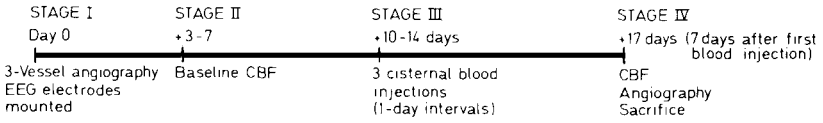



Fig. 1. Experimental protocol for SAH model in baboons, including the various steps in the evaluation of cerebrovascular changes.

Mayfield aneurysm clip. Branches of the external carotid artery were ligated to further eliminate errors due to isotope contamination from extracerebral tissue. Measurements of baseline CBF were then performed. At the end of the session the linguo-facial trunk was ligated and the clip on the external carotid artery removed.

*Stage III (cisternal blood injections).* Three to 7 days after the baseline CBF measurements, when the animals had recovered, they were sedated with phencyclidine and anesthetized via a face mask with fluothane in a concentration of about 1%. The animals were allowed to breathe spontaneously during the injection procedure. The head was lowered by tilting the animal 20°. Blood injections were usually carried out 3 times with 1-day intervals. Fresh autologous blood was obtained from a femoral artery puncture. The first and second injections were given into the cisterna magna and the third through the orbit into the cisterna chiasmatis. A short-beveled needle (1.5 mm o.d.) with a stylet was passed percutaneously into the cisterna magna, and 5–10 ml CSF was withdrawn and replaced with blood until the respiratory rhythm became depressed. This usually occurred after administration of 1–4 ml more than the amount of CSF aspirated. For the injections into the chiasmatic cistern, the same type of needle was passed through the conjunctiva and then through the optic foramen in a lateral direction in order to avoid puncture of the third ventricle. Three to 6 ml of blood was injected; it was not possible to aspirate more than 1–2 ml CSF prior to the injection. The total amount of blood injected in the 13 surviving animals is given in Table I.

*Stage IV.* The animals were again prepared for angiography and CBF measurements 7 days after the first blood injection. After 3-vessel angiography, the catheter for injection of $^{133}Xe$ was inserted into the right external carotid artery. Three to 4 stable baseline CBF measurements were performed before starting the actual experiments. At the end of the entire experiment the animal was killed by exsanguination while still anesthetized. Thirteen animals, listed in Table I, followed the complete experimental protocol as outlined in Fig. 1.

The brain was removed immediately and the circle of Willis and adjacent major branches were carefully dissected out with the aid of an operating microscope. Arterial segments were prepared for pharmacological tests of the reactivity to noradrenaline and 5-hydroxytryptamine in vitro[105] and for electron microscopy (to be published). The brains were sectioned for gross pathological examination.

*Physiological parameters*

Body temperature was continuously monitored by a rectal thermometer and was maintained between 36.5 and 37.5 °C using a heating blanket under the animal. A femoral artery and vein were catheterized prior to the CBF measurements. Arterial blood pressure was measured continuously, and blood samples were withdrawn for the determination of pH, $p_aCO_2$, $p_aO_2$ (pH-Blood Gas Analyzer 413, Scandiametric, Sweden), total arterial oxygen content (Eschweiler, F.R.G.), and hemoglobin concentration. Further, the superior sagittal sinus was catheterized for continuous pressure registration (SSP) and measurement of venous oxygen content. The catheters were withdrawn after each experimental session. The femoral vessels were sutured using microsurgical technique, and the sagittal sinus defect was covered with hemostatic cellulose cotton (Oxycel, Parke-Davies). All procedures were carried out under sterile conditions.

*EEG recording*

The EEG activity was recorded with 3 bipolar leads from 4 electrodes inserted into the skull bone parasagittally on each side of the head from frontal to occipital position, using an Elema-Schönander electroencephalograph (time constant, 0.3 s; upper frequency limit, 70 Hz), before and during each CBF measurement. The EEG pattern was classified by a

clinical neurophysiologist according to the scale adapted from the description of Domino[26].

*CBF measurements*

CBF was measured using the intra-arterial technique[62,63,74]. $^{133}Xe$ (4 Bq) was dissolved in 0.9% saline to a final volume of 0.5 ml and was injected manually into the right internal carotid artery via a catheter placed in the linguo-facial artery. The injection time was close to 1.5 s, and one operator always performed this injection to ensure reproducibility. Ipsilateral extracerebral contamination of $^{133}Xe$ was minimized by clamping the external carotid artery (in addition to the previously described extirpation of the temporal muscle).

The clearance rate for $^{133}Xe$ was measured from the temporo-parietal area of the brain during a 10-min period with a single, collimated sodium iodide scintillation crystal of 25 mm diameter and 15 mm in collimator depth. Correct and reproducible placement of the scintillation crystal was achieved by fixing the head of the animal and the crystal in a stereotaxic frame.

Pulses from the detector were amplified and passed into an analyzer, which accepted pulses equivalent to an energy range of 70–90 keV. The output from the analyzer was recorded by a scaler and a ratemeter (Model 30-32 analyzer; Selektronik, Denmark) and registered by a potentiometer writer. The initial counting rate after the isotope injection was kept between 300 and 900 cps.

The stochastic (height over area) method was used for calculating CBF. The following equation was applied:

$$\text{CBF} = 100\,\lambda\,\frac{\text{height}}{\text{area}}\ \text{ml/100 g/min},$$

where $\lambda$ denotes the mean tissue/blood partition coefficient for $^{133}Xe$ corrected for differences in the hemoglobin concentration. Height designates the initial height of the curve minus the background. Area is the total number of impulses recorded during 10 min minus the background.

The $p_aCO_2$ was measured during the 3rd and 7th min, respectively, of the 10-min $^{133}Xe$ clearance period. The average value was maintained near 40 mm Hg and was not allowed to exceed the range of 38–42 mm Hg. The $p_aCO_2$ was kept between 100 and 150 mm Hg. The $CMRO_2$ value was obtained by multiplying the arterio-venous difference of oxygen content with CBF. It should be noted that the blood taken for $CMRO_2$ determinations was drawn from the superior sagittal sinus, which only drains part of the brain region in which flow was measured.

*Gross neuropathological examination*

Examinations were performed on all of the 24 blood-injected animals. Seven animals died after the blood injections and were studied within a day after death, whereas the remaining animals were examined following the final CBF measurement; these were sacrified by exsanguination during deep fluothane anesthesia. The skull was opened, the brain was removed and examined for the presence of blood in the subarachnoid space, for gross external changes, and for internal pathology on sliced tissue.

*Angiography*

Selective angiography of both carotid arteries and one vertebral artery was performed ad modum Seldinger, following percutaneous puncture of the femoral artery in the groin. A standard transfemoral catheter-set for children with preformed S-shaped catheter tip was used (Radicath Cerebral Child; Radiplast, Sweden; 1.14 mm i.d., and 1.57 mm o.d.). The arterial blood gases were monitored before each vessel examination: $p_aCO_2$ was kept within the range of 38–42 mm Hg and $p_aO_2$ between 100 and 150 mm Hg.

Each vascular territory was examined in anteroposterior and lateral projections (for the first projection, 6 films were exposed at a rate of 2/s and for the lateral view, 5 films at 2/s were followed by 3 films at 1.5-s intervals). For each series, 2.5 ml of metrizamide (Amipaque; Nyegaard, Norway) in a concentration of 280 mg iodine/ml was manually injected during approximately 1 s. This contrast medium was chosen because of its relative inertness; it is non-ionic, its osmolarity is one-third that of conventional contrast media, and it has lower acute toxicity and less vasodilatory effect than other media[3]. All angiographies on each animal were performed by the same radiologist to ensure reproducibility. The duration of the entire angiographic procedure was normally 30–45 min.

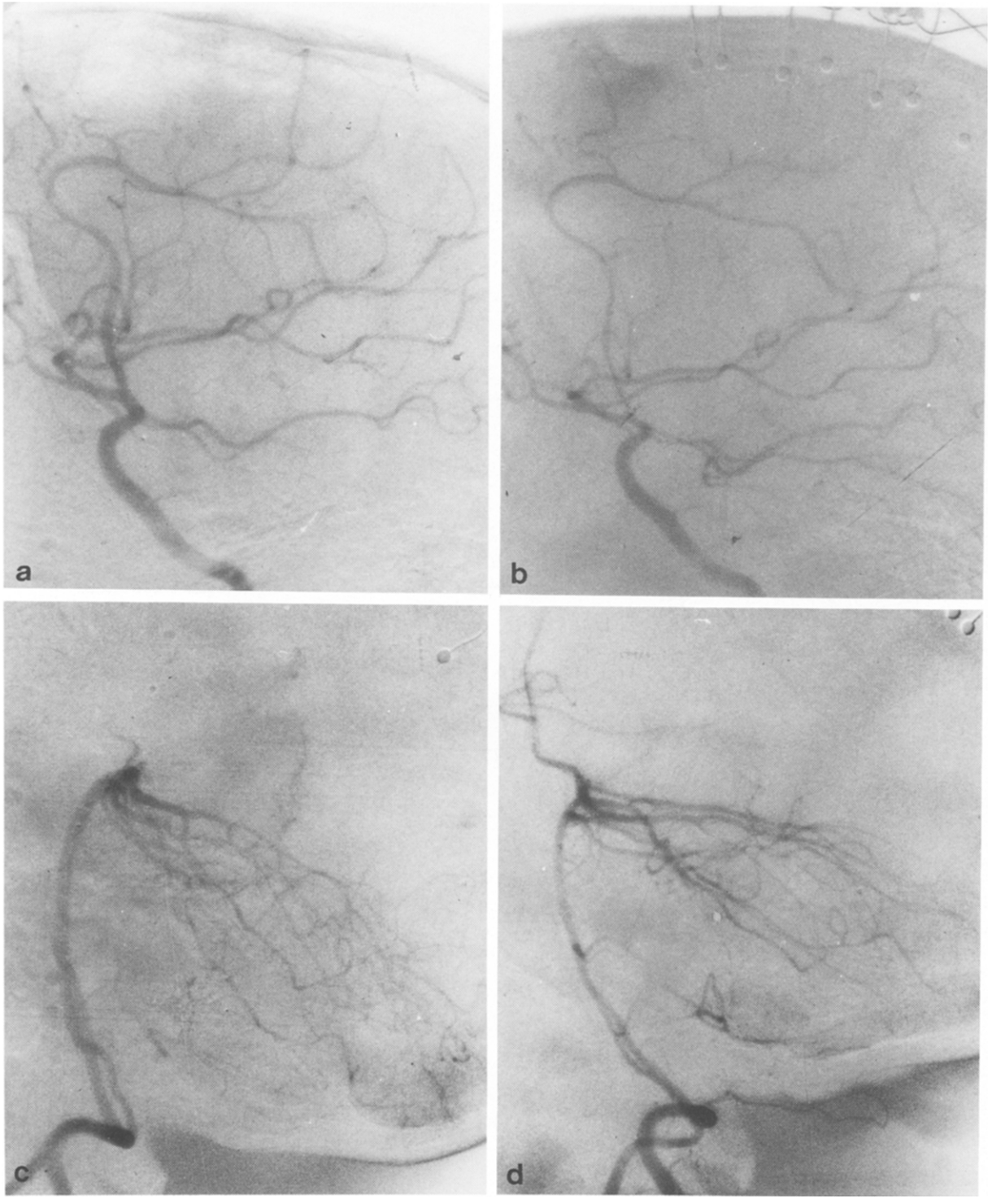



Fig. 2. Cerebral angiograms in lateral projections of the carotid (a and b) and vertebro-basilar (c and d) systems before (a and c) and one week after (b and d) experimental SAH. In the carotid angiograms the degree of vasospasm (b) was −31% (Table II, animal No. 3) as compared to the control situation before cisternal blood injection (a). A generalized vasoconstriction can be seen in all the major cerebral arteries of the carotid system, including the proximal part of truncus anterior. In the vertebro-basilar angiograms the spasm (d) was −29% (Table II, animal No. 4) in comparison with the control (c).

*Determination of arterial caliber*

Measurements of arterial diameters before and after subarachnoid blood injections were performed with a technique similar to that described by Gabrielsen and Greitz[47]. For each vessel examined, the anterior and lateral films showing optimal arterial filling (Fig. 2) were chosen for measurements as indicated in Fig. 3. The contour of the vessel was enhanced with a sharp soft pencil with the aid of a magnifying glass. The diameter was measured with an attempted accuracy of 0.05 mm using a calibrated precision lens with fixed magnification and lens-to-film distance. Values from 5 different measuring points were averaged in each of the right and left carotid examinations, and the mean of 8 values was used for the vertebro-basilar system.

Measurements were made in random order without previous knowledge of the status of the animal or whether or not blood had been injected. Some of the films were measured on two separate occasions without any statistically significant difference in the determined average values. In order to evaluate the total variations caused by possible differences in anesthesia and acid–base status, by spontaneous intra-individual variations, as well as by variations in measurements, 2 animals were examined on 3 different occasions each (twice on one day and once 1–2 weeks later). The standard deviation of the means amounted to 2%, the maximal difference being 5% ($n = 18$).

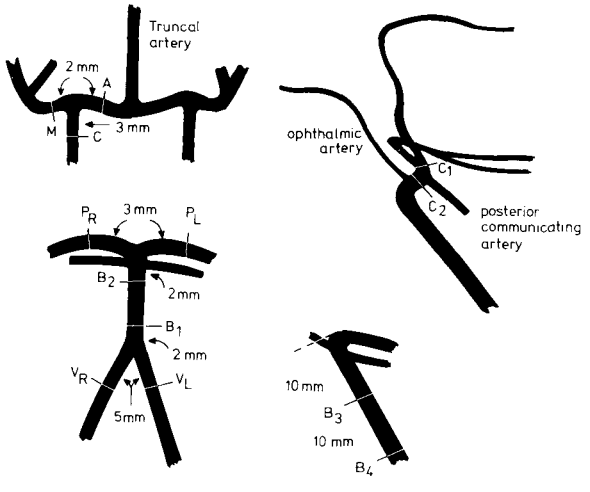



Fig. 3. Measuring points to obtain values for arterial diameters (see Table II) on carotid (left) and vertebral (right) angiograms (see Fig. 2). M, middle cerebral artery; A, anterior cerebral artery; C, internal carotid artery in antero-posterior view; $C_1$, internal carotid artery distal to origin of posterior communicating artery; $C_2$, internal carotid artery distal to origin of ophthalmic artery; $V_R + V_L$, right and left vertebral arteries; $B_{1-4}$, basilar artery; $P_R + P_L$, right and left posterior cerebral arteries. Distances from well-defined landmarks are shown.

The term 'vasospasm' is commonly used to describe various degrees of angiographically visible vasoconstriction following SAH in patients. Clinically, the vasoconstriction shows a variety of angiographic appearance. It may comprise a general narrowing, it may be segmental, or have a beaded appearance. In animal experiments, vasospasm is often used to designate any degree of angiographically well-defined vasoconstrictory response following various attempts to mimic SAH in man.

*Induction of hypercapnia and hypertension*

The CBF response to hypercapnia was tested in 8 control animals and 7 SAH animals one week following experimental SAH. Three of these animals were tested both before and after cisternal blood injection. Hypercapnia was induced by adding approximately 5% $CO_2$ to the $N_2O/O_2$ gas mixture. A stable level of $p_aCO_2$ was obtained within 30 min. The hypercapnic response in CBF was calculated by comparing the mean CBF value of 2–5 determinations during normocapnia with the flow value obtained during the maximum level of $p_aCO_2$ tested.

The CBF response to hypertension was tested in 7 control animals and 7 animals belonging to the SAH group in order to evaluate the autoregulatory capacity. Two of the animals were studied both before and after the experimental SAH. Angiotensin II (Hypertensin, Ciba) was infused intravenously in order to achieve approximately 40–50 mm Hg increase in MABP above normal. A 0.5 mg/ml stock solution of the agent was diluted with 0.9% saline to a varying degree (which differed in the individual animals, particularly with regard to development of tachyphylaxis) to obtain the desired level of hypertension. Attempts were made to keep the MABP steady for 5 min before and throughout the CBF measurement. In 2 of the controls and in 3 of the SAH animals the test for CBF autoregulation was performed after the previously described hypercapnic response, which was followed by normalization of the physiological parameters during 30–60 min.

*Administration of nimodipine*

The action of the calcium entry blocker, nimodipine, was studied in two types of experiments: one in order to test its therapeutic effect on CBF following experimental SAH, and one with the aim of evaluating its influence on blood flow autoregulation before and after cisternal blood injections.

The nimodipine preparation for clinical application (Bayer AG) was used. The concentration of the antagonist in that preparation is 0.1 mg/ml in a solution of distilled water, containing 150 mg/ml each of ethanol and polyethyelene glycol. Nimodipine was omitted from the solution given in control experiments. Both solutions were, immediately before administration, diluted with 0.9% saline to a suitable volume for infusion (approximately 0.5 ml/min), which took place at room temperature. Syringes and catheters were wrapped in aluminium foil in order to avoid photodecomposition of nimodipine.

In the therapeutic experiment, nimodipine was tested in 7 animals before and (the same animals) after experimental SAH. The blocker was infused intravenously at a rate of 0.5 $\mu$g/kg/min, and CBF, $CMRO_2$ as well as physiological parameters were registered at 45 min after starting the infusion. The same volume of solvent (vehicle) alone was infused in the control experiments. In the SAH animals, the effect of nimodipine was evaluated 7 days after the first cisternal blood injection.

The autoregulation experiment was carried out on 5 animals before and 5 after experimental SAH, at 7 days following the first blood injection. Three of the animals were tested both before and after the injections. The CBF was measured after infusion of angiotensin II into a different peripheral vein than that used for administration of nimodipine. An MABP level approximately 30–40 mm Hg above normotension was attempted.

## RESULTS

*General effects of cisternal blood injection*

*Mortality and morbidity.* Of the 24 animals receiving cisternal blood injections, 7 died before the final angiography and the blood flow measurement could be performed. They all showed signs of cerebral edema and tentorial herniation at autopsy. In addition, one of these animals had a subdural and one an intra-

TABLE I

*General neuropathological data, including the amount of blood found in the basal cisterns and the ventricular system, following experimental SAH (volume of blood injected is given) in 13 animals going through the complete experimental protocol*

| *Animal No.* | *Body weight (kg)* | *Sex* | *Blood injected* (ml)* | *Cisternal blood*** | *Intraventricular blood**** | *Intracerebral hematoma* | *Miscellaneous* | *Neurological deficits* |
|---|---|---|---|---|---|---|---|---|
| 1 | 14.0 | M | 17.0 | 0 | Yes (I–IV) | No | – | No |
| 2 | 9.2 | M | 14.0 | ++ | No | No | – | Drowsy, ataxia, progressive hemiparesis dx |
| 3 | 13.3 | M | 22.5 | +++ | Yes (I–IV) | No | Bilat. incipient tentorial herniation | Drowsy, general ataxia |
| 4 | 14.1 | M | 19.5 | +++ | No | No | Brain edema | No |
| 5 | 9.4 | F | 19.5 | ++ | Yes (I–IV) | No | Hydrocephalus | No |
| 6 | 16.6 | F | 14.5 | + | No | No | – | No |
| 7 | 11.8 | M | 18.0 | ++ | Yes (III–IV) | Thalamus bilat. | Brain edema | Hemiparesis sin |
| 8 | 15.0 | M | 26.5 | +++ | No | No | – | No |
| 9 | 13.4 | M | 33.0 | ++ | No | No | – | No |
| 10 | 12.2 | M | 14.5 | + | No | No | – | No |
| 11 | 12.3 | M | 16.0 | ++ | No | Caps int dx | Right-sided edema, incipient herniation dx | No |
| 12 | 20.5 | M | 19.0 | ++ | No | No | – | No |
| 13 | 21.8 | M | 18.0 | ++ | Yes (III–IV) | No | – | Drowsy |

* Total amount of blood injected into cisterna magna and chiasmatis.

** Thickness of clotted subarachnoid blood coat was estimated: + designates a layer less than 1 mm thickness, ++ corresponds to 2–5 mm, and +++ above 5 mm thickness.

*** The cerebral ventricles are numbered from I to IV.

cerebral hematoma. Four of the remaining 17 animals that underwent angiography were discarded for various reasons (two had an occlusion of the intracavernous part of the right internal carotid artery, one showed occlusion of the right superior cerebellar artery, and the fourth animal developed signs of tentorial herniation during the CBF measurements and was found to have a brain abscess at autopsy). Thus, the mortality rate due to the experimental SAH was 29%. The pattern of morbidity is summarized in Table I.

*Gross neuropathological findings.* Varying amounts of blood were found in the cistern systems of all animals except one (No. 1 in Table I). In addition, blood was found in the ventricles of 5 animals (No. 1, 3, 5, 7, and 13). In animal No. 7, autopsy revealed a big clot trapped in the dilated third ventricle showing periventricular hemorrhagic foci with necrosis. This inadvertent deposition of blood during one of the injections was probably responsible for the edema and accompanying paresis. The fresh hemorrhage in the internal capsule of animal No. 11 may have been the result of a very recent infarction from an embolus via a branch of the middle cerebral artery, probably during withdrawal of the catheter.

The extent of neurological deficits is also illustrated in Table I. A right-sided hemiparesis was recorded 2 days after the last blood injection in animal No. 2. The paresis and additional ataxia progressed during the subsequent 24 h, until the final CBF examination. This animal showed a particularly high degree of angiographically visible vasospasm (Table II).

*Physiological parameters.* Animals subjected to experimental SAH did not differ significantly from controls in their mean levels of $p_aCO_2$, MABP, or pulse rate (Table III). The SSP was increased following SAH in 7 out of the 10 animals in which complete pressure recordings could be obtained. The mean hemoglobin concentration was reduced after the experimental SAH (Table III).

TABLE II

*Percent decrease (negative values) or increase (positive values) in arterial diameter following cisternal blood injections. Angiograms were compared before the blood injections and one week after the first injection (cf. Fig. 1)*

The measuring points for obtaining the values of arterial diameter are shown in Fig. 3. The numerical values obtained for the carotid system were $(A+M+C+C_1+C_2)/5$ and for the vertebro-basilar system $(V_L+V_R+B_1+B_2+B_3+B_4+P_L+P_R)/8$.

| *Animal No.* | *Total amount of blood injected (ml)* | *Right carotid system* | *Left carotid system* | *Vertebro-basilar system* |
|---|---|---|---|---|
| 1 | 17.0 | + 5 | + 3 | |
| 2 | 14.0 | + 5 | −44 | −22 |
| 3 | 22.5 | −31 | −21 | −25 |
| 4 | 19.5 | − 6 | − 8 | −29 |
| 5 | 19.5 | −11 | −10 | − 9 |
| 6 | 14.5 | −15 | −19 | −34 |
| 7 | 18.0 | −17 | −26 | −23 |
| 8 | 26.5 | − 5 | −30 | −21 |
| 9 | 33.0 | −25 | −17 | −28 |
| 10 | 14.5 | − 9 | − 5 | −28 |
| 11 | 16.0 | − 8 | −18 | −14 |
| 12 | 19.0 | −11 | −20 | −23 |
| 13 | 18.0 | −15 | −18 | |
| Mean value | 19.4 | −11** | −18*** | −23*** |

Arterial dimensions pre-SAH vs post-SAH according to Student's *t*-test: $^{***}P < 0.001$; $^{**}0.001 < P < 0.01$.

*Effect of experimental SAH on CBF and $CMRO_2$*

The mean baseline CBF value of the 13 animals studied was 63.2 ml/100 g/min (Table IV). The values obtained after correction for a $p_aCO_2$ of 40 mm Hg are also shown in the table. The flow was reduced by 18% one week after experimental SAH; the reduction was slightly more pronounced when the flow had been corrected to $p_aCO_2$ of 40 mm Hg. Also the brain oxygen metabolism was lowered by 11%, following the subarachnoid blood injections. Corrections to a $p_aCO_2$ of 40 mm Hg potentiated the difference in $CMRO_2$ in comparison with the situation before experimental SAH (Table IV). All changes were statistically significant.

*Changes in arterial diameter following experimental SAH*

The percent change in vascular diameter, measured in the carotid and vertebro-basilar systems, following experimental SAH in comparison with dimensions before blood injection is summarized in Table II. Repeated cisternal injections of blood resulted in a statistically significant reduction of mean arterial diameter in all vascular territories measured one week after the first injection (Fig. 3). The degree of vascular narrowing was more pronounced in the vertebro-basilar than in the carotid system. It should be recalled that the largest amount of blood to simulate

TABLE III

*Physiological parameters for different experimental groups, before and after experimental SAH*

Values are means ± S.E.M.; $n$ = number of experiments. Besides the intentionally elevated levels of $p_aCO_2$ and MABP in hypercapnia and hypertension, respectively, there were no significant differences between mean values either between the different groups of experiments or between pre- and post-SAH animals, except for pulse rate and SSP which were significantly increased after SAH as compared to control values, while the hemoglobin (Hb) concentration was lower after SAH as compared to pre-SAH values.

| *Experimental condition* | *n* | *$p_aCO_2$ (mm Hg)* | *MABP (mm Hg)* | *Pulse rate (beats/min)* | *SSP (mm Hg)* | *Hb conc. (g/l)* |
|---|---|---|---|---|---|---|
| *Before experimental SAH* | | | | | | |
| Basic model | 13 | 39.2 ± 0.3 | 111 ± 5 | 89 ± 4 | 2.9 ± 1.0[a] | 112.5 ± 3.7 |
| Normocapnia | 8 | 39.3 ± 0.3 | 106 ± 3 | 81 ± 11 | 2.9 ± 2.0 | – |
| Hypercapnia | 7 | 57.5 ± 2.0 | 110 ± 6 | 104 ± 7 | 17.1 ± 5.6* | – |
| Normotension | 7 | 39.5 ± 0.2 | 111 ± 4 | 97 ± 1 | 4.1 ± 1.8 | – |
| Hypertension | 7 | 39.8 ± 0.5 | 115 ± 2 | 96 ± 7 | 4.2 ± 1.6 | – |
| *After experimental SAH* | | | | | | |
| Basic model | 13 | 39.9 ± 0.3 | 110 ± 3 | 108 ± 7* | 9.6 ± 3.0[b,*] | 99.8 ± 4.9** |
| Normocapnia | 7 | 40.0 ± 0.5 | 112 ± 4 | 102 ± 10 | 5.0 ± 2.0 | – |
| Hypercapnia | 7 | 55.9 ± 1.6 | 111 ± 6 | 108 ± 10 | 16.7 ± 3.3 | – |
| Normotension | 6 | 39.9 ± 0.4 | 106 ± 3 | 103 ± 8 | 6.5 ± 3.3 | – |
| Hypertension | 6 | 41.4 ± 0.3 | 150 ± 2 | 107 ± 10 | 9.3 ± 3.1 | – |

In Student's $t$-test: $^{**}0.001 < P < 0.01$, $^{*}0.01 < P < 0.05$.
[a] Only 12 animals.
[b] Only 10 animals.

SAH was deposited in cisterna magna. There was in the individual case no direct relationship between the total volume of blood injected and the degree of reduced vascular diameter. Although 8 of the transorbital blood injections were performed on the right side, there was no statistically significant side-to-side difference in diameter of the carotid system.

*Effect of experimental SAH on CBF response to hypercapnia*

The mean CBF value for the normocapnic control animals was 64.1 ± 5.1 ml/100 g/min (Table V). During hypercapnia, the flow increased with a mean of 3.90 ± 0.75 ml for each mm Hg increase in $p_aCO_2$. In the individual experiments, the relative increase in flow was more prominent at the lower level of hypercapnia than at higher $p_aCO_2$ values, in accordance with observations by Boisvert et al.[17]. The mean CBF in the normocapnic animals following SAH was 57.2 ± 3.6 ml/100 g/min (Table V). Hypercapnia induced a flow enhancement that was less pronounced than before SAH, amounting to 1.72 ± 0.32 ml for each unit increase in $p_aCO_2$. The hypercapnic CBF response was thus significantly impaired following SAH (comparison of hypercapnic indices: $P = 0.02$). The mean values for $CMRO_2$ were not affected by hypercapnia either before or after experimental SAH.

The mean values for the physiological parameters in the experiments with hypercapnia are summarized in Table III. Before experimental SAH, hypercapnia was usually associated with a slight increase in MABP of the individual animals, though the mean values were not statistically different from those of the normocapnic animals. All hypercapnic animals had elevated pulse rate and SSP. One week following SAH, hypercapnia induced similar changes in pulse rate and SSP as in the control animals. The mean MABP remained unaltered (Table III).

*Effect of experimental SAH on CBF response to hypertension*

In the control animals not subjected to experimental SAH the induced hypertension did not significantly affect CBF (Table VI). The pulse rate in the individual animals usually decreased initially, followed by tachycardia when the hypertension had been established. The changes were not sufficient to be reflected in the mean pulse rate. The $p_aCO_2$ levels remained unaltered (Table III).

Following experimental SAH, the autoregulation

TABLE IV

*Changes in CBF and $CMRO_2$ one week following experimental SAH, expressed both as actual values and following correction to $p_aCO_2 = 40$ mm Hg*

Values are means ± S.E.M. $n$ = number of experiments.

| *Experimental group* | *n* | *CBF (ml/100 g/min)* | *$CMRO_2$ (ml/100 g/min)* | *CBF corr. (ml/100 g/min)* | *$CMRO_2$ corr. (ml/100 g/min)* |
|---|---|---|---|---|---|
| Before SAH | 13 | 63.2 ± 4.4 | 3.22 ± 0.13 | 66.1 ± 4.4 | 3.37 ± 0.12 |
| After SAH | 13 | 51.6 ± 2.8** | 2.88 ± 0.08** | 51.8 ± 2.8** | 2.88 ± 0.11** |

Student's $t$-test: $^{**}0.001 < P < 0.01$.

was impaired in 5 of the animals as evidenced by a pressure-dependent increment in CBF during hypertension. The remaining 2 animals in this experiment showed a preserved autoregulatory response. The mean increase in CBF ($P = 0.02$ compared to controls) during the elevated MABP is summarized in Table VI. The recorded physiological parameters behaved similarly in the post-SAH as in the pre-SAH animals (Table III).

Calculation of the autoregulatory index, describing the amount of flow increase in relation to the degree of elevation of systemic blood pressure (range, see Table VI), gave 0.06 ± 0.01 ml/100 g/min for each mm Hg (mean ± S.E.M.) in the pre-SAH animals and 0.29 ± 0.03 ml/100 g/min per mm Hg post-SAH ($P < 0.001$).

### *Effect of intravenous nimodipine on CBF and $CMRO_2$ before and after experimental SAH*

Infusion of the calcium antagonist, nimodipine, at a rate of 0.5 μg/kg/min intravenously caused a statistically significant, approximately 17%, increase in CBF before experimental SAH (Fig. 4). This was associated with an elevation also in the rate of oxygen consumption. The $CMRO_2$ response was not statistically significant, however, even when taking into

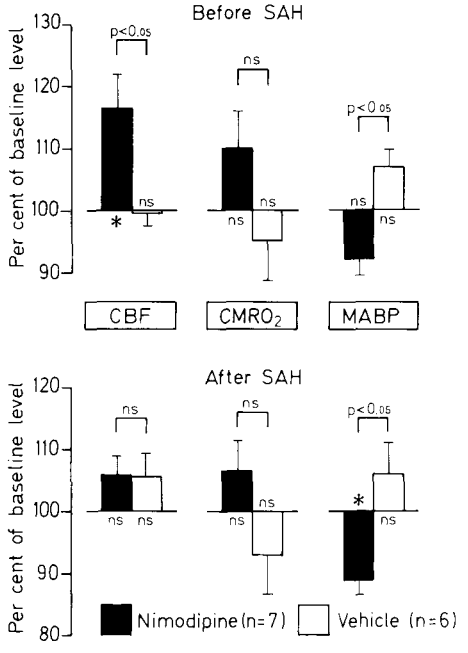



Fig. 4. Effect on CBF, $CMRO_2$ and MABP measured 45 min after a continuous i.v. infusion of nimodipine (0.5 μg/kg/min) or vehicle (at a corresponding infusion rate), before and after experimental SAH (the same 7 animals were tested before and after cisternal blood injections). Significance levels are shown for the comparison between active solution and vehicle ($t$-test for unpaired observations) as well as between baseline and post-infusion levels (paired $t$-test): $^{*}0.01 < P < 0.05$, ns = non-significant difference.

TABLE V

*Hypercapnic response of CBF and $CMRO_2$ before and one week after cisternal blood injection*

Values are means ± S.E.M., $n$ = number of experiments.

| *Experimental group* | *n* | *Normocapnia* | | *Hypercapnia* | |
|---|---|---|---|---|---|
| | | *CBF (ml/100 g/min)* | *$CMRO_2$ (ml/100 g/min)* | *CBF (ml/100 g/min)* | *$CMRO_2$ (ml/100 g/min)* |
| Before SAH | 8 | 64.1 ± 5.1 | 3.67 ± 0.15 | 126.3 ± 7.5 | 3.52 ± 0.37 |
| After SAH | 7 | 57.2 ± 3.6 | 2.99 ± 0.14 | 86.2 ± 3.6 | 3.05 ± 0.17 |

TABLE VI

*Ability of CBF to autoregulate (and effects on $CMRO_2$) during angiotensin-induced hypertension at the MABP levels indicated, in comparison with control levels of normotension tested*

Values are means ± S.E.M., $n$ = number of experiments.

| *Experimental group* | *n* | *Normotension* | | | *Hypertension* | | |
|---|---|---|---|---|---|---|---|
| | | *MABP (mm Hg)* | *CBF (ml/100 g/min)* | *$CMRO_2$ (ml/100 g/min)* | *MABP (mm Hg)* | *CBF (ml/100 g/min)* | *$CMRO_2$ (ml/100 g/min)* |
| Before SAH | 7 | 100–120 | 57.1 ± 1.9 | 3.42 ± 0.12 | 150–160 | 59.5 ± 2.1 | 2.86 ± 0.23 |
| After SAH | 7 | 95–115 | 52.0 ± 2.0 | 2.92 ± 0.14 | 140–160 | 65.6 ± 4.2 | 2.90 ± 0.27 |

Paired $t$-test showed no significant difference between normotensive and hypertensive flow values before experimental SAH, but a significant difference with $P = 0.02$ after SAH. The difference in $CMRO_2$ (before SAH) did not reach statistical significance.

consideration the influence of the vehicle. The nimodipine infusion caused a decrease in blood pressure (see also Table VIII). The effect of nimodipine was principally the same in animals subjected to experimental SAH, though the CBF response was smaller and did not reach statistical significance.

*Effect of intravenous nimodipine on CBF autoregulation before and after experimental SAH*

Administration of nimodipine (0.5 μg/kg/min i.v.) during the increase in MABP induced by intravenous angiotensin II resulted in a marked elevation of CBF in the pre-SAH animals, amounting to a mean (± S.E.M.) increase of 1.58 ± 0.33 ml/100 g/min for each mm Hg of pressure increment. The corresponding effect of the hypertension was significantly less pronounced ($P < 0.02$) following experimental SAH: 0.58 ± 0.13 ml/100 g/min per mm Hg increase in MABP. The flow differences between the normotensive and hypertensive states are summarized in Table VII. As can be seen from the blood pressure ranges in Tables VI and VII, it was not possible to obtain the same high degree of hypertension in the presence of nimodipine as compared to the situation without calcium antagonist, either before or after experimental SAH. Table VII also shows that the impaired autoregulation was not accompanied by any significant alteration of brain oxygen metabolism.

In order to elucidate the effect of the vehicle on CBF autoregulation in these experiments, the same volume of solvent without nimodipine was infused as above in 6 experiments on pre-SAH animals. The CBF was not significantly changed during the induced hypertension compared to the normotensive situation 45 min after the infusion; in fact, mean flow values were identical.

In three of the pre-SAH animals an abrupt increment in SSP by approximately 20 mm Hg was noted at the highest levels of hypertension (MABP close to 140 mm Hg). The mean elevation in the pressure was from 1.4 to 12.0 mm Hg in the whole group of nimodipine-treated animals before experimental SAH

TABLE VII

*Effect of intravenous nimodipine infusion (0.5 μg/kg/min) on CBF autoregulation and $CMRO_2$ during angiotensin-induced hypertension at the MABP levels indicated, before and after experimental SAH*

Values are means ± S.E.M., $n$ = number of experiments.

| *Experimental group* | *n* | *Nimodipine during normotension* | | | *Nimodipine during hypertension* | | |
|---|---|---|---|---|---|---|---|
| | | *MABP (mm Hg)* | *CBF (ml/100 g/min)* | *$CMRO_2$ (ml/100 g/min)* | *MABP (mm Hg)* | *CBF (ml/100 g/min)* | *$CMRO_2$ (ml/100 g/min)* |
| Before SAH | 5 | 90–120 | 58.3 ± 2.7 | 3.42 ± 0.29 | 120–145 | 94.8 ± 11.0* | 3.18 ± 0.25 |
| After SAH | 5 | 95–115 | 54.6 ± 1.6 | 3.00 ± 0.12 | 125–140 | 70.8 ± 2.2** | 3.01 ± 0.44 |

Paired $t$-test showed significant differences between CBF values at normotension and hypertension: *$0.01 < P < 0.05$, **$0.001 < P < 0.01$. There were no statistically significant differences in the $CMRO_2$ values.

TABLE VIII

*Physiological parameters for the various groups in the nimodipine experiments, before and after experimental SAH (cf. Fig. 4 and Table VII)*

Values are means ± S.E.M., $n$ = number of experiments. Nimodipine significantly lowered MABP (see also Fig. 4), and tachycardia as well as elevated SSP were seen during the hypertensive challenge (not evident in the absence of nimodipine, see Table III). Differences in other physiological parameters were not statistically significant.

| *Experimental condition* | *Before experimental SAH* | | | | | *After experimental SAH* | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | *n* | $p_aCO_2$ *(mm Hg)* | *MABP (mm Hg)* | *Pulse rate (beats/min)* | *SSP (mm Hg)* | *n* | $p_aCO_2$ *(mm Hg)* | *MABP (mm Hg)* | *Pulse rate (beats/min)* | *SSP (mm Hg)* |
| Nimodipine | 7 | 39.7 ± 0.3 | 92 ± 3 | 109 ± 4 | 4.9 ± 1.3 | 7 | 39.7 ± 0.5 | 89 ± 3 | 110 ± 3 | 6.3 ± 2.0 |
| Vehicle | 6 | 39.0 ± 0.3 | 107 ± 3 | 113 ± 9 | 4.0 ± 2.3 | 6 | 39.5 ± 0.4 | 106 ± 5 | 107 ± 7 | 6.0 ± 6.0 |
| Nimodipine during normotension | 5 | 39.9 ± 0.3 | 110 ± 6 | 100 ± 9 | 1.4 ± 0.8 | 5 | 39.9 ± 0.72 | 103 ± 4 | 99 ± 6 | 3.9 ± 2.0 |
| Nimodipine during hypertension | 5 | 40.7 ± 0.3 | 137 ± 5 | 131 ± 9 | 12.0 ± 4.1 | 5 | 41.8 ± 0.9 | 136 ± 4 | 122 ± 9 | 10.0 ± 3.7 |

(Table VIII), whereas little or no change was seen in the hypertensive animals (pre- and post-SAH) not receiving the antagonist (Table III). The corresponding differences in SSP were less evident in the nimodipine-treated animals following experimental SAH (Table VIII).

## DISCUSSION

### *SAH model*

Spontaneous SAH in man often originates from ruptured aneurysms mostly situated in the circle of Willis. The rupture results in a massive bleeding into the cisterns and the subarachnoid space. The spread of blood seems to be highly dependent on the configuration of the cisterns, which show a great variation in individual patients[132]. The blood clot may be trapped and confined to one or two cisternal compartments related to the ruptured aneurysm, though the bleeding more often involves several compartments together with blood contaminating the whole subarachnoid space.

Experimental SAH has been produced in different ways: by damaging intracranial arteries, by injection of blood into the cisternal systems, or by direct deposition of blood around major arteries through a craniotomy. The resulting vascular reaction appears to involve a mechanical factor, particularly when an artery is torn to induce the bleeding[31,32,65,90,102,109,131]; (for further references, see Heros et al.[61]; Mohr and Kase[86]). The effect of experimental SAH has been evaluated by several methods, e.g. direct inspection usually through an operation microscope[9,28,87], angiography[25,30,31,109], measurement of CBF[24,31,92,110], or by recording metabolic changes in the brain[35,36,38] recently on the basis of emission tomography[77,82].

It has often been observed that the vasospasm following SAH has a biphasic time-course (e.g. Heros et al.[61]). A primary spasm occurs immediately after the appearance of blood in the subarachnoid space and abates after 1–3 h. The vasospasm reappears after 2–4 days to reach a maximum by approximately one week. This late spasm coincides with the clinically deleterious phase following the SAH.

The present model in baboons represents an effort to create a reproducible late type of spasm with a decreased CBF and measurable signs of brain dysfunction. The vasoconstrictory changes were induced with repeated, massive intracisternal blood injections, and the model was designed to be used in therapeutic experiments with a calcium entry blocker.

By repeated blood injections over a one-month period, Weir et al.[129] were able to produce vasospasm which, however, did not last more than a few hours and which was associated with a weekly mortality of 6–37%. Simone et al.[110] produced a biphasic type of spasm, the late phase extending over 3 days after the blood injection in about one-third of the monkeys. In a small series of baboons, du Boulay et al.[27] found angiographically visible late vasospasm in 2 out of 5 animals receiving blood in the cisterna magna. On the other hand, Pickard et al.[94] and Boisvert et al.[17] did not see any vasoconstrictory reaction in angiographic examinations on the 7th day after a sin-

gle injection of autologous blood in an amount of 0.75 ml/kg into the suprachiasmatic cistern of baboons. Following injection of twice as much blood in this cistern of cynomolgus monkeys, Espinosa et al.[31] reported angiographically visible cerebral vasospasm in the course of 3 weeks after the experimental SAH. The degree of vascular reaction was found to be related to the amount of simulated SAH apparent on computerized tomography. When a 7-ml fresh blood clot was placed in contact with the major arteries of the anterior basal circulation through a craniotomy[32], the degree of vasospasm was even higher, with 31–100% reduction in vessel calibre in 87% of the animals subjected to angiography. In an SAH model in the dog, Brawley et al.[20] and Kuwayama et al.[72] demonstrated a biphasic vasospasm, and repeated blood injections (4 ml autologous blood twice with a one-day interval) have successfully been used in this animal to produce angiographically demonstrable vasoconstriction during subacute or chronic phases following the experimental SAH[125]. In the rat a maximal late vasoconstriction was found on day 2 (ref. 25) and in the rabbit at 3–5 days after cisternal blood injection[30].

In order to cover the late phase of spasm as completely as possible, the present animals were studied one week after the first blood injection and 3 days after the last injection. The repeated injections were given with the particular intention to obliterate the cisternal systems in order to mimic the human situation in animals having a more open cisternal system. Weir et al.[129] introduced the technique of repeated cisternal blood injections by performing weekly administrations over a period of 4 weeks in monkeys. The overall mortality rate was 75%, and the vasospasm was limited to a few hours; late spasm did not occur. The mortality rate from the cisternal blood injections was 29% in the present series of baboons. All of the 13 animals except one undergoing angiography showed a varying degree of vasospasm, ranging between 5% and 44% of the vessel calibre measured before experimental SAH. The angiographic distribution and the time-course of manifestation of the vasoconstriction much resembled the human situation following spontaneous SAH[128]. However, a segmental type of spasm, like that often seen in patients, was never observed. It has been claimed that this latter type indicates a resolution phase and that its appearance hence depends on the time at which the angiographic examination is performed after the bleeding[61,83,84,86].

There was no correlation between the degree of angiographically visible vasoconstriction and the total volume of blood injected. This may, however, be a less representative correlate than the amount of locally accumulated blood (cf. Fischer et al.[41]; Espinosa et al.[31]), which should be related to the degree of trapping of the blood in the cistern systems.

For practical reasons it is more difficult to evaluate neurological symptoms in surviving monkeys than in patients. Thus, the less impressive degree of neurological deficiencies in our monkeys does not necessarily indicate a major difference between the SAH model and the human situation with regard to the degree of brain damage.

*Anesthesia*

The animals were maintained on a basal anesthesia with 70% nitrous oxide and 30% oxygen. During the surgical procedures, fluothane was added to the gas mixture, and during CBF measurements and angiography the animals were given intravenous infusion of phencyclidine. The rate of infusion was modified according to the EEG in order to maintain as far as possible stable activity corresponding to Stage 1 and Stage 2 of Brierly et al.[21]. Although there are different opinions as to the effect of nitrous oxide on CBF and $CMRO_2$, most studies favour the view that its influence is minimal (for references, see Siesjö[108]). Several groups of investigators have reported values for CBF and $CMRO_2$ in primates given phencyclidine as supplementary anesthesia that are similar to those of non-anesthetized primates and even to man[44,51,95,99,100].

*CBF and metabolism following experimental SAH*

The validity of the $^{133}Xe$ method for CBF measurements requires that the isotope is distributed to and detected from the brain only. In order to avoid tracer contamination by the extracerebral tissue, all branches of the external carotid artery were ligated and the temporal muscle on the ipsilateral side was extirpated[42,74]. The stochastic (height over area) method was used to calculate CBF since, without computerization, it provides a more reliable assessment of the $^{133}Xe$ clearance than the initial slope and

the compartmental methods. Moreover, any occasional cortical excitation during phencyclidine administration will have less influence on the CBF measurements.

Our observation that there was a decrease in CBF in the 'post-hemorrhagic period' is in accordance with reports from several clinical investigations[12,39,52,59,64,82,89,114]. A decrease in CBF has been reported in animal studies shortly after experimental SAH[55,92,110]. In the present series of experiments, reduced CBF was also demonstrated during a time period corresponding to the late phase of vasospasm. In addition, an associated decrease in $CMRO_2$ was found. A similar reduction in the metabolism has been described in clinical studies by Grubb et al.[52] and Martin et al.[77], and in animal experiments by Fein[35], though only in the acute stage of experimental SAH. The fall in CBF and $CMRO_2$ cannot be explained simply by an increase in intracranial pressure (cf. Fein[35]), since the SSP was only elevated in two-thirds of the animals without direct correlation to alterations in flow and metabolism.

*$CO_2$ response after experimental SAH*

It was observed in patients with acute SAH[71] that the response of CBF to hypercapnia was impaired, while the reaction to hyperventilation was preserved. In experimental acute SAH, failing CBF response to changes in $p_aCO_2$ has been demonstrated[27,56,93]. The hypercapnic reaction is impaired also a week after intracisternal blood injection in baboons[17]. In their latter series, the response to increased respiratory carbon dioxide was 1.6 ml/100 g/min per mm Hg of elevation in $p_aCO_2$, as compared to 3.27 ml/100 g/min per mm Hg in an untreated control group, thus very similar to the present results. It is notable that none of the animals in the study by Boisvert et al.[17] showed angiographically visible signs of vasospasm, whereas all except one animal in the present series showed vascular narrowing. Thus, it appears that vasoconstriction is not a prerequisite for a reduced flow response to hypercapnia.

It was first reported by Pickard and MacKenzie[95] that the prostaglandin synthesis inhibitor, indomethacin, abolishes the CBF reaction to hypercapnia in normal baboons, without affecting the autoregulatory mechanism. The observations have been confirmed in studies on rats[23,107]. Since the circulatory response to hypoxia, on the other hand, was not affected by indomethacin in these experiments it is not conceivable that the arachidonic acid metabolites in question represent general coupling substances for the chemical control of CBF.

Although there is a direct relationship between the ability of the cerebrovascular bed to respond to increased $p_aCO_2$ and the cerebral oxygen consumption of the brain[46], the slight (and non-significant) reduction in $CMRO_2$ following experimental SAH in the present study is not sufficient to explain the curtailed carbon dioxide response. The lower control CBF in the SAH animals, together with the slightly elevated mean SSP, might have had an influence in terms of a lower cerebral perfusion pressure. However, it should be recognized that a reduced circulatory response to hypercapnia is present in experimental SAH also under conditions when control CBF (in normocapnia) is similar to that recorded before the SAH[17].

*CBF autoregulation after experimental SAH*

The cerebral circulation has the capacity to maintain its flow in the face of changes, such as arterial pressure, that would tend to alter the flow[70]. It is evident from clinical studies that SAH in an unpredictable number of patients results in an impairment of this blood flow autoregulation, as tested during either hypotensive or hypertensive conditions[34,49,59,64,88,96,113,133]. The consistency of the impairment is difficult to evaluate because the materials presented are small, the tests are not uniform and, moreover, have not been performed at comparative time intervals following the bleeding.

Animal experiments carried out during the acute phase following subarachnoid blood injection have shown damage of the autoregulatory response[16,37,56]. However, the acute results are not necessarily applicable to the chronic situation in patients. In studies attempting to simulate the clinically important chronic phase following SAH, Pickard et al.[94] found an impaired ability to maintain a constant CBF, both during hypotensive and hypertensive tests carried out one week after the injection. In contrast to the present findings, however, angiographic evidence for vasospasm was not found.

The impaired CBF autoregulation expresses itself either as a passive increase in flow during elevation of

systemic blood pressure or, if CBF is maintained relatively constant, in an earlier break-through of the autoregulation at high pressure. The variability in the type and degree of impairment may reflect that this not only involves a functional injury to the autoregulatory mechanism, including the perivascular sympathetic nerves[14,29,42], but also may have a structural component, whereby the brain vessels to a varying degree have lost the ability to alter their calibre. Changes in the function of perivascular sympathetic nerves have been observed in association with subarachnoid blood[30,45,90], and structural alterations in the vessel wall have been reported under such conditions (for references, see Espinosa et al.[33]).

Induced hypertension represents one of the few widely used therapeutic approaches having positive effects on the neurological symptoms following SAH[69]. Owing to the risk for brain edema due to the damaged autoregulation, this treatment can obviously be advocated only during strict control of intracranial pressure. Recently, there has been an increasing interest in predicting the risk for the development of late ischemia following SAH. The autoregulatory response has been one of the parameters used for this purpose[34,96].

*Nimodipine and CBF autoregulation*

Contractile responses of blood vessels depend on calcium derived from several sources, intracellular as well as extracellular[15,63,124]. Nimodipine is one of several calcium entry blockers currently studied in vascular pharmacology. This antagonist has a preferential action on cerebral and coronary blood vessels[58,67,119,122], probably because these vascular beds are more dependent on extracellular calcium for their contractile process[2,78].

Intravenous infusion of the commercially available nimodipine solution impaired CBF autoregulation, particularly in animals studied before experimental SAH. The less pronounced effect following SAH may be related to the structural[22,33,91,115,118] or functional[97,105] changes in the vessel wall observed in conjunction with the chronic phase of vasospasm, allowing for a less marked vasodilatation to occur in response to the induced hypertension. It should be noted that large-calibre arteries, such as those represented in the pial vascular beds where the above-mentioned structural and functional changes have been observed, contribute more to total resistance in the cerebral circulation than in other vascular beds[10].

It has recently been reported from experiments on baboons that nimodipine impairs the autoregulatory response to reduced blood pressure[54,79]. In the present study, intravenous infusion of the same calcium antagonist was found to affect also the autoregulatory response to induced hypertension. Using an equal dose of nimodipine, Harris et al.[54] did not note any effect on the hypertensive response. The discrepancy may be related to the mode of inducing the hypertension where direct cerebrovascular effects of metaraminol may have compensated for the impairment by local vasoconstriction to a greater degree than angiotensin II used as vasopressor agent in the present study.

The cellular mechanisms underlying the blood flow autoregulation are not established. It appears that the myogenic tone developing in response to stretch of the vascular wall[11] plays an important role. It has also been shown that the perivascular sympathetic nerves are able to markedly modulate the autoregulation, studied both at its lower limit[42,43,60] and its upper limit[14,29,80]. The myogenic tone as well as the neurogenic vascular tone induced by noradrenaline both require extracellular calcium sources which are sensitive to calcium antagonists (e.g. Bevan[13]). Therefore, the effect of nimodipine on autoregulation would be predictable, whether its action is intracellular or at the level of the sarcolemmal calcium channels[5]. This is also consistent with the observations of an impaired CBF response to changes in arterial carbon dioxide levels following nimodipine administration[54,79].

In order to exclude that the effects on autoregulation were due to the ethanol component of the commercially available nimodipine solution utilized, control infusions were made with the same concentration and amount of ethanol in the group of animals where the effect on autoregulation was most pronounced (pre-SAH group). In these control experiments infusion of ethanol at a rate of 0.75 mg/kg/min did not affect the autoregulatory response to induced hypertension. In accordance with this, McQueen et al.[81] found loss of autoregulation only in 8 out of 15 monkeys with an intravenous infusion dose as high as 1200 mg/kg/min, and autoregulation was not abolished in all animals until the dose was doubled. When

the commercially available nimodipine solution is infused intravenously for therapeutic purposes into patients at the same rate as that used in our baboons, the ethanol concentration in blood does not even reach measurable levels[76].

*Therapeutic trial with nimodipine*

Against the background of the central role of calcium ions in the contractile mechanisms of vascular smooth muscle[111] it was logical to develop calcium antagonists with a preferential site of action in the cerebrovascular bed[121] for therapeutic application in the vasospasm associated with SAH. The calcium entry blocker, nimodipine, is a potent dilator of pial arteries[6,8,18,105,117,119,122] and it induces a varying degree of CBF increase[53,54,58,67,78,85,104].

In the present series of experiments, CBF in the pre-SAH animals was increased by 17% following intravenous infusion of nimodipine at the rate of 0.5 $\mu$g/kg/min. This is the same order of magnitude as reported in monkeys by others (Harper et al.[53]: 27% at 2 $\mu$g/kg/min; Harris et al.[54]: 5–24% at 0.6 $\mu$g/kg/min; McCalden et al.[79]: 18% at 1 $\mu$g/kg/min). Following experimental SAH the mean flow enhancement with nimodipine was not distinguishable from the control value in animals receiving the vehicle. Both before and after experimental SAH there was a (non-significant) tendency to an increased $CMRO_2$ following nimodipine treatment.

The flow response agrees with observations in isolated pial arteries removed from baboons subjected to cisternal blood injection, showing a less pronounced effect of nimodipine compared to results with vessels from pre-SAH animals[105]. The difference in the nimodipine effects pre- and post-SAH may be related to the probably impaired ability of vessels in the chronic phase of spasm to dilate efficiently, as previously discussed. In acute spasm, on the other hand, local application of calcium uptake blockers has pronounced cerebrovascular dilatory effect, as shown in dog models[75,126]. It should be noted, though, that the mode of administration is different; that the CBF response to calcium uptake blockade is characteristically less pronounced in monkeys and man (see also refs. 8, 53, 54, 79) than in laboratory animals such as rat[85], rabbit[58], cat and dog[6,67]; and that associated metabolic vasodilatation cannot be excluded. The difference between the pre- and post-SAH monkeys was not evident following continuous intracarotid infusion of nimodipine[104].

The effect of the vehicle on CBF and $CMRO_2$ was varying, slight, and always statistically non-significant. There is no reason to assume that ethanol should have similar cerebrovascular effects as nimodipine since, if anything, ethanol has a vasoconstrictory action in the brain[4,101].

Several attempts have been made to apply nimodipine clinically in order to overcome the neurological deficits due to delayed cerebral ischemia secondary to cerebral vasospasm in conjunction with SAH in patients. In a multi-center study Allen et al.[1] indicated an improvement of the angiographic appearance of the cerebrovascular bed with nimodipine, but this has not been possible to confirm in other studies[7,50,76]. In spite of an unchanged degree of angiographically visible vasospasm, all these authors have reported less ischemic sequelae following treatment with the calcium antagonist. The improvement might be due to dilatation of smaller cerebral arteries or arterioles not visible on angiography. It has been shown that small-calibre arteries are more sensitive to calcium entry blockers than larger vessels[19,117]. Since nimodipine infusion has little[104] or no (present study) influence on CBF following experimental SAH (in contrast to its more substantial vasodilatory action under normal conditions, as already discussed) the beneficial effect may instead be exerted at the neuronal level by a direct prevention effect on ischemic cell death (e.g. ref. 123). In fact, regarding the presently demonstrated tendency to impairment of CBF autoregulation by nimodipine following experimental SAH, it would rather seem that nimodipine might have an unfavourable influence on the cerebral circulation under this pathological condition in primates.

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